

Student: Jonatan Rabjerg
Date: 1/15/24

Instructor: Andreas Basse-O'Connor
Course: Calculus Beta

Assignment: MyLab Math - Uge 11

Find the unique solution of the second-order initial value problem.

$$y'' - 20y' + 100y = 0, y(0) = 2, y'(0) = 26$$

$y = e^{rx}$ is a solution to the second-order homogeneous differential equation of the form $ay'' + by' + cy = 0$ if and only if r is a solution to $ar^2 + br + c = 0$. The equation $ar^2 + br + c = 0$ is called the auxiliary equation to $ay'' + by' + cy = 0$.

Write the auxiliary equation for $y'' - 20y' + 100y = 0$.

$$r^2 - 20r + 100 = 0$$

Solve the auxiliary equation.

$$r^2 - 20r + 100 = 0$$

$$(r - 10)(r - 10) = 0 \quad \text{Factor.}$$

$$r = 10 \quad \text{Solve.}$$

The auxiliary equation has one real repeated root. The general solution to a second-order homogeneous differential equation of this type is shown below.

$$y(x) = c_1 e^{rx} + c_2 x e^{rx}$$

Therefore, the general solution to $y'' - 20y' + 100y = 0$ is as follows.

$$y(x) = c_1 e^{10x} + c_2 x e^{10x}$$

To determine the particular solution for the given initial conditions, first find y' , or the derivative of $y = c_1 e^{10x} + c_2 x e^{10x}$ with respect to x .

$$y'(x) = c_1 (10e^{10x}) + c_2 (10x e^{10x} + e^{10x})$$

Now, substitute 0 for x , 2 for $y(0)$, and 26 for $y'(0)$ in the resulting expressions for y and y' , and then simplify to find the values of c_1 and c_2 under the given initial conditions.

$$y(x) = c_1 e^{10x} + c_2 x e^{10x}$$

$$y(0) = c_1 e^{10 \cdot 0} + c_2 (0) e^{10 \cdot 0}$$

$$2 = c_1$$

$$y'(x) = c_1 (10e^{10x}) + c_2 (10x e^{10x} + e^{10x})$$

$$y'(0) = 10c_1 e^{10 \cdot 0} + 10c_2 (0) e^{10 \cdot 0} + c_2 e^{10 \cdot 0}$$

$$26 = 10c_1 + c_2$$

Solve the second equation for c_2 , substituting 2 for c_1 .

$$26 = 10c_1 + c_2$$

$$26 = 10(2) + c_2$$

$$6 = c_2$$

Finally, substitute the values of c_1 and c_2 into the general form of the solution to find the particular solution of $y'' - 20y' + 100y = 0$ when $y(0) = 2$ and $y'(0) = 26$.

$$y(x) = c_1 e^{10x} + c_2 x e^{10x}$$

$$y(x) = 2e^{10x} + 6x e^{10x}$$